

ZOO507 Solved Past Paper

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Regards VUWAYS Team

Welcome To All New Members

507 Solved Past Papers

Q1: Write down the component of ecology?

Two main component of ecosystem are:

- i- Abiotic: non living elements: air, current, minerals
- ii- Biotic: living part of ecosystem: animal, plants, fungi, bacteria
- iii- Consumers in chain
- iv- Decomposers & nutrient cycle-

Q2: Sub-fields of biology?

Chemistry:- It is the study of ~~existing~~ chemicals-

Sub fields:- Ecology:- organisms interact with one another & with their non living things-

Palaeontology:- study of fossils & ancient life

Anthropology:- study of human culture

Hydrology:- study of earth resources i.e.: lakes, oceans

Climatology:- earth's atmosphere & its climatic changes

Botany:- plants Zoology:- animals

Q3: Define Environmental Sciences?

A: It is an interdisciplinary study of connection b/w earth's life support system & human activities-

Q4: Define Environmentalism?

a social movement dedicated to protecting the earth's life support system for us & for others-

Q5: Environmental Sciences fields?

Anthropology

Demography

Economics

Political Sciences

Humanities

(Q. 6)

Q. 6: Environmental Sustainability?

A sustainability is the ability of earth's various natural system & human cultural system & economics to survive & adopt the changing environmental condition ~~also~~ indefinitely.

Q. 7: Define Sustainability?

A cultural component of sustainability is natural resources & natural service that keep us & other forms of life alive & support our economic -

Q. 8: Define Nutrient cycling?

A vital science is nutrient cycling "the circulation of chemical necessary for life life form, environment (mostly from air & soil) through organism & back to environment."

Q. 9: Define & example of prokaryotes?

They are one-celled organism that do not have a nucleus or any other membrane-bound organelles inside them. e.g. bacteria & cyanobacteria

Q. 10: How solar/light energy to chemical energy by plants?

Solar energy converted to chemical energy by plants is known as photosynthesis -

→ The process by which green plants & some other organisms use sunlight to synthesize nutrients from CO_2 & water.

Q: ¹¹ How green plants use solar energy?
In plants, photosynthesis involves the green pigment chlorophyll & generates oxygen as by-product.

Q: ¹² Types of energy?
light energy, heat energy, gravitational --- chemical energy, sound, chemical, atomic energy. Each form can be converted or changed into other form.

Q: ¹³ Base of energy quality?
Energy quality is contrast b/w diff forms of energy, the diff trophic levels in ecological system & the propensity of energy to convert from one to another.

Q: ¹⁴ What is natural capital?
It is the world's stock of natural resources, which includes geology, soil, air, water & all living organisms. that human derive a wide range of services, often called ecosystem services in which makes human life possible.

Q: ¹⁵ Natural capital uses & degradation?

- 1- It is the environmental stock or resources of Earth that provide goods, ²⁻ flows, ³⁻ ecological services required to support life
- 4- It has financial value as the use of natural
- ⁵ derive many businesses.
- 8- It is the better use & care of natural resources.

Q¹⁶ 6) Degradation: (natural capital degradation) involves using resources at an unsustainable rate. Means, They are wasting the natural resources like burning of coal at extreme level.

Q¹⁷ Effects of human on our environment?
Human impact on our environment or anthropogenic impact on the environment includes changes to biophysical environment and ecosystem, biodiversity loss, and natural resources caused directly or indirectly by humans, including global warming, environmental degradation, mass extinction of resources, over consumption, over exploitation, pollution and deforestation. These are all the result from human activity.

Q¹⁸ How ecological footprints are growing?

The world-average ecological footprint in 2014 was 2.8 global hectares per person. The carbon footprint is the fastest growing part of ecological footprint and accounts currently for about 60% of humanity's total ecological footprint.

Imp
of
atmos
phere

1. Its very small fraction weigh on earth
2. Its imp bcz its highly reactive
3. Circulates & plays imp role evolution of land scape.

Q¹⁹ The mixture of gases that surround earth called atmosphere. Composition of atmosphere in ecosystem?

4. Nitrogen accounts for 78% of atmosphere
Oxygen 21% & Argon 0.9%. Gases like carbon-dioxide, nitrous oxide, methane and ozone are trace gases that about the tenth of one percent of atmosphere.

Q²⁰ Factor assembly community?

It describes how the first species to colonize an ecosystem determine what end state of that ecosystem look like. The accumulation of species diversity over time in habitat.

5

Cultural eutrophication is the process that speeds up natural eutrophication bcz of human activity. Due to clearing of land & buildings of towns & cities, land runoff is accelerated & more nutrient such as phosphate & nitrate is spread to lakes & rivers, then to coastal estuaries & bays.

Q²² Why water H_2O is Important?

Water is one of the most imp substance on earth. All plants & animal must have water to survive. If there is no water there would be no life on earth. Apart from drinking to survive it is use for many other works - It keeps body temperature moderate, lubricate joints.

Q. 23 Composition of geosphere?

The three main layers of earth includes the ^{thinnest} crust (1% of the earth's volume) the thick (mostly rock) mantle (84%) & the core (inner & outer combined, 15%)

Q²⁴ Major factors that support life on earth?

Earth has basic requirements are a source of energy - protection from UV, ^{liquid solvent} water, gases. Ideally placed in our solar system & galaxy to support life.

Q²⁵ Five example of tertiary consumers?

Tertiary consumers can be carnivores.
They can eat ^{of other carnivores} meat. ~~to~~ hawk, wolves, Tiger, leopard, snakes.
Basically, they are predators feed on primary consumers.

Q26: Give example of secondary consumers?

feed on herbivores: cows, zebras, rabbits, deer -

Q²⁶ Imp of diatoms? Q

Mark ND factors of diatom?
Unicellular algae, shells having interlocking parts. Major producer of ^{system} aquatic

Dinoflagellate:- unicellular, double flagell and imp producer of an aquatic form

Q²⁷ Define Omnivorous? (primary consumers)

Can eat everything: Includes pigs, foxes, cockroaches & Humans. Dentition:- These teeth are
• ripping • chiseling • grinding

Q²⁸ Detritivorous feeders?

They feed on wastes or dead bodies, known as detritivorous feeders. e.g. mites, earthworms
• catfish • nullies

Q²⁹ Define Decomposers?

Decomposers break down dead organisms & feces by digesting organic matter extracellularly & absorbing the product of digestion. They feed by secreting enzymes e.g. bacteria & fungi

Q³⁰ Features of phytoplankton?

Microscopic organisms that float or drift in water. They are a great source of food for aquatic organisms. Like: whales, shrimp, snails. They are 2 types: Diatom, Dinoflagellate

Q³¹ Define Red tide?

The harmful & toxic bloom of dinoflagellate causes the colour of sea water to turn red. Eg. in Suez Nile Egypt.

Q³² Food web?

In natural ecosystem, most consumer feed on more than one type of organism, & most organism are eaten or decomposed by more than one type of consumer. bcz of this, organism in most ecosystem form a complex network of interconnected food chains called food web. It shows how producer, consumer & decomposer are connected to one another as flow of energy through trophic level in ecosystem provide stability in ecosystem.

Q³³ Food chain?

The sequence of organism through which energy moves in an ecosystem called food chain.
plant → insect → frog → hawk
decomposers use died hawk & return their nutrient to reuse of by Producer.

Q³⁴ Food web?

When an abiotic factor become out of an organism tolerance range then it is said to be limiting factor - e.g. Oxygen & water^{food}

Q³⁵ Taxis?

Response of an organism towards any abiotic factor is called Taxis. an organism orient itself to response e.g. response to light called phototaxis.

Q³⁶ Trophic level?

Food web is complex, so its easy to group organism according to the form of energy used. These group called -
- consumer; - primary, secondary, tertiary -
- omnivore, carnivore, herbivore

Q: ³⁷ Producers :- photoautotroph :- chemoautotroph
 producers are autotroph (self feeders) ^{nutrient & energy} obtained by environment
 2 types :- photoautotroph & They capture sunlight
 to produce energy rich carbohydrate ^{glucose} by photosynthesis
 chemoautotroph
 Bacteria convert ^{inorganic} energy through complex ^{nutrient} reaction
 compounds by using hydrogen sulphide instead of
 sunlight called chemosynthesis

Q: ³⁸ Ecological Niche & role of bears in ecosystem?
 Ecological niche is the role of animal in our
 ecosystem. It is defined as the role that species plays in
 environment including behaviour & influence. It's a way of life
 in a community that includes everything that affect its survival & reproduction -
 • activity pattern: remain active • actual place of living
 • resources gain from surrounding environment. ^{role of bears in ecosystem}
 • lives in forest • feeds on nuts, blood sucking insects & parasites -
 • carry seed of berries in stomach for long distance & expel in new place
 to grow in new area • hibernates in winter season.

Q: ³⁹ How living system are sustained? •
 All living systems from cell to biosphere are dynamic
 system that are constantly changing in response
 to changing environmental conditions -

* continent movement * climate changing
 * Natural disturbance * Succession

Living system contain complex networks of positive
 & negative feedback loops that interact to provide
 some degree of stability, or sustainability, extended life span.

Q: ⁴⁰ Factors of carrying capacity?

The maximum population of a living species that a particular
 habitat can sustain without being degraded - The carrying
 capacity of an area or volume is not fixed -
 availability • water • space • environment condition

Q⁴¹: Define weathering? process of decomposition & disintegration that causes the rocks to break down in natural way i.e. chemical, mechanical, biological weathering

Q⁴²: Earth's life support system?

Earth's life support system consists of four main physical spherical systems that interact with one another

- atmosphere (air)
- hydrosphere (water)
- geosphere (rocks, soil)
- biosphere (living things)

Q⁴³: Importance of coral reef?

- corals are tiny living animals that belong to a group of organism called Cnidarians. They are two main types of coral reefs - soft corals & hard corals.
- coral reef produced by large mass of calcium carbonate that secreted by coral in marine water.
- They are an amazing center of aquatic biodiversity.

Benefits:-

- provide natural habitat
- Economic benefit
- food
- Medicine
- Recreation
- breeding ground
- Nursing ground.

Q⁴⁴: Diff b/w rocky & sandy shores?

The area of shoreline b/w low & high tide is called intertidal zone.

* Rocky shores :- They face a huge impact of waves, so large no. of species occupy diff. niche in intertidal zone in response to following changing environmental conditions.

- Temp.
- water flow
- salinity

* Sandy shores :- Sandy beaches & their adjoining coastal wetlands are a home to a variety of shore birds that feed in specialized niche on crustaceans, insects & other organisms.

Organism can survive by adopting following adaptations

- burrowing in sands
- Digging in sand
- Tunnelling in sands.

Q⁴⁵ Soil Importance: • Nutrient • water • habitat
• Seed germination • Biological productivity

Q⁴⁶ Soil Type: can be determined by • Topography
• climate • Time length -
In general diff b/w Two type of Soil

1 -> Pedalfon Soils:

- Acidic Soil formed in wet temp.
- Extensive leaching activity
- Support growth of deciduous forest & tall grasses -

2- Pedacoll Soil: (acid & semid)

- Soil that formed in dry climate & basic in nature
- less leaching in activity
- promotes growth of desert shrubs & small / medium grasses.

↓ Types

3- Laterite Soils:

a special type of pedacoll that is bright red & form in Tropical & Subtropical climate with heavy rain & high Temp.

4- Caliche: A tough cemented layer developed by dense mass of calcium carbonate.

Q⁴⁷ Soil Horizon:-

A vertical section of soil that tells about various layers of soil as zones are named as Soil horizon.

It has 4 Zones: O zone:- uppermost zone of soil where leaves & fallen & other plant or matter deposited.

- A zone Zone of leaching they accumulate in B zone
- B zone Zone of accumulation they accumulate in C zone
- C:- composed of broken bedrock material that merged with entirely solid & hard bedrock material.

Q: 48 Temperature?

Temperature controls metabolic rate as well as overall activity of an individual.

Human: 98.6°F

goat: 103°F

bird: 105°F

Warm blooded animal :- these animal that maintain their internal body temp. constant. Birds / mammals / ^{homeotherms}

Cold blooded animal :- animal whose internal body temp. changes accordance to external envt. Temp. also called Poikilotherms. amphibians / fish / reptiles

Q: 49 Imp of Temperature?

~~metabolism~~

~~growth~~

~~reproduction~~

- Metabolism
- Seed germination
- opening & closing flowers
- Seasonal falling of ^{leaves} trees
- Growth & development
- animal migration due to seasonal changes in temp.
- reproduction
- Temp. = (hot day & cold night) in desert.

Q: 50 Define Thermoregulation?

ability of an organism to regulate its body temp.

- Metabolism
- Growth
- reproduction

Influenced by

- Availability of food
- water
- shelter

→ Optimum temp. :- Enzyme or biological catalyst needs a specific temperature.

Improper temp. makes the enzyme unable to perform their catalytic activity.

Q: Write down the concept of a species?

Ans: A group of sexually reproducing organisms that can mate & produce fertile offspring. This is biological concept of species.

- Every organism is a member of a particular species with specific trait.

→ Biological concept of species:-

This defines species on the basis of their interbreeding potential not on similarities or appearance. e.g. western meadowlark & eastern meadowlarks look similar but they do not interbreed with each other, so they are separate species.

→ Ecological concept of species:-

It defines species in terms of its ecological niche.

→ Morphological concept of species:-

It defines species by its structural features.

→ Phylogenetic species concept:-

Define species as the smallest group of individuals on a phylogenetic tree.

Q: Population:- A group of same species located at the same time is called population.

Q: Population characteristics:-

population size

population disturbance

population growth

population density

Q: Population limits:- If environment is bad place to stay i.e. it must be too hot, too cold, or too dangerous.

Geographical limit:-

it also a main constraint that decrease the size of population.

Q⁵⁵: Community c. is the group of population of diff species located in a particular area or place.

- Communities are not only random composition of species but they also have a unique organization.

Q⁵⁶: Community assemblage:-

The community is an assemblage of ~~or more~~ species populations that occur together in space & time.

Q⁵⁷: Community assembly:- The species that assemble to make up a community are determined by.

- dispersal constraint
- Environmental constraint
- Internal dynamics

Q⁵⁸: Biotic components:-
 producer & green plants
 consumer: animal
 Decomposer & bacteria & fungi

Q⁵⁹: Generalist species c. are the species which have broad range of niches are known as generalist species.

- they can live in many ^{diff} places, eat variety.
- often tolerate a wide range of environmental conditions.
- Eg:- cockroaches, mice, rats, racoon & human

Cockroaches present over 350 million species

- high reproduction rate
- eat variety of food
- can survive after 48 to 60 days of freeze
- resistant to poison
- plays role in food web, tasty meal for birds & insects.
- about 25 species, locate carry 25 diff viruses & bacteria ^{illness}

Q⁶⁰: Specialist species c.

Those species that occupy narrow range of niche are known as specialist species.

- They are more prone to extinction-
- They may be able to live only in one habitat
- Use one or few type of food
- Tolerate a narrow range of climate & Env. changes

eg. tiger Salamanders :- breeds only in fishless ponds where their larva will not be eaten

- They show reduce competition
- They show sharing of limiting resources

Q: ⁶¹ Is it better to be a generalist or specialist?
Specialist has an advantage bcz they have few competitors but under rapidly changing environmental conditions, the generalist can adapt quickly so, they can survive it depends on the present conditions.

Q: ⁶¹ Native Species are those species that normally live & thrive in a particular ecosystem bcz of its soil, isolation & climatic conditions-

- They occur naturally within a region-
- They either involve in that region or arise & become established in that region without human assistance-

Endemic species These species are found only in a specific part of the world only. eg. Indian dolphin in India only

Indigenous species: They are found within the region & elsewhere in the world also. eg. Red kangaroo only in Australia only.

Q: ⁶² Non-native species-

Species that are non-native to their env. & are deliberately or accidentally introduced into an ecosystem & called non-native species-

- Also named as invasive • alien • exotic species
- Invasive species are those who are harmful to humans or other communities

Q⁶³ Impact of Non-Native species?

- Non-native species spread rapidly if find new, more favourable niche
- In new niches these species often do not face the predators & disease they face.
- They may be able to out-compete some native species in their new niche
- Causes habitat loss & ecosystem disturbance

Q⁶⁴ Indicator Species?

Species that provides early warning of damage to a community or an ecosystem called Indicator species.

- e.g. The presence of ^{or absence} trout species in water at Temp. within their range of tolerance is an indicator of water quality, as they need clean water with high level of oxygenated -

Q⁶⁵ Keystone Species?

Some species plays a key & crucial role in maintaining the structure of a biological community - known as keystone species - They have large effect on types & abundance of other species in an ecosystem

- e.g. - Sharks also a keystone species bcz they remove injured, sick animals from ocean, thus they play an imp. role.

Q⁶⁶ Foundation Species?

Good e.g. of beaver make dams on rivers & other waterways to make paths for other species.
Plays a major role in shaping communities - They create & Enhance the habitat in ways that provides benefits to other species - They facilitate the survival & reproduction of other species in community.
• e.g. - Elephants push over, break up, uproot trees, grasses, shrubs in grasslands & woodland of Africa which helps other species.

Q⁶⁷: Ecological pyramids ?

An ecological pyramid is graphical relationship among diff trophic structures or level in an ecosystem - • trophic means diff levels of organism. 3 main types.

→ Pyramid no.s - represents the no of individuals at each trophic level of an ecosystem.

→ Pyramid biomass - Indicates mass of living matter at each trophic level of food chain

→ pyramid of energy - determines the flow of energy through each trophic level in food chain

Q⁶⁸: Food Pyramid ?

It is a graphical representation of flow of energy through diff trophic level of eco.

(Population (v. imp))

Q⁶⁹: Population Dynamics ?

It is a study of how these characteristics of population change in response to changes in environmental conditions : Environmental conditions are :

- Temperature
- harmful chemical
- resources availability
- driving or dispersing of competing spe

Q⁷⁰: Population ^{distribution} dispersion: 1. clumping
2. uniform clumping 3. random clumping

Q: ¹¹ Population factors: Population differs in factors such as

- Distribution
- Numbers

- edge structures
- Density

17

Q: ¹² clumping:-

Individuals found in form of groups or clumps due to uneven distribution of resources-

Q: ¹³ Population / uniform dispersion:-

the tendency for population to be found evenly distributed about their habitat

- This is caused by a species ability to survive anywhere in their habitat

Q: ¹⁴ Random / Population dispersion:-

distribution of population in random fashion through out their habitat generally, this type is rare while clumping distribution is common-

Q: ¹⁵ Population growth:-

drinal population change over time as result of

- Birth
- Death
- Dispersal

→ Biotic potential: ability of population to grow under ideal condition

→ low biotic potential Large individual, such as elephant & blue whale

→ high biotic potential Small individual, such as bacteria & insects

Q:¹⁶ Intrinsic rate of increase:- the rate at which population of specie would grow if it had limited resources.

- response early life
- short generation time
- reproduce many times
- how many offspring may live

Q:¹⁷ Environmental resistance:-

the constraints that climate, food, space, & other environmental factors place on a population called Env. res. -

Q:¹⁸ carrying capacity K ?

It is combination of all factors that act to limit the growth of a population. Together, biotic potential & environmental resistance determines the carrying capacity K .

Q:¹⁹ Survivorship curves: plots of No. of survivor. Type 1 curves live out their potential life span.

- " 2 " " mortality rates are constant throughout age class.
- " 3 " " mortality rate is higher for youngest cohorts.

Q:²⁰ Define Exponential growth?

A rapid increase in no. of indiv of a population at a constant rate of growth.

→ Exponential or geometric growth starts slowly but then accelerates as the population increases, bcz the base of size is increases.

Q.⁸¹ Population Age Structure :- has a strong effect on population growth. Populⁿ usually discussed with:-

- The pre-reproductive age :- not mature enough to reproduce
- reproductive age :- Those capable to reproduce
- post reproductive age :- Those too old to reproduce

Q.⁸² Reproductive potential :- Ability of individual to reproduce under ideal or optimum condition. Exponential growth cannot occur indefinitely.

Q.⁸³ Factors of reproductive Influence.

- No. of offspring produced
- The likelihood of survival to reproductive age
- Duration of reproductive period.
- Length of time it takes to reach maturity.

Q.⁸⁴ Logistic population growth?

Logistic growth involves rapid exponential potential growth followed by a steady decrease in population growth until population size is level off.

Q.⁸⁵ Population regulation:-

Demography :- Study of human population & Factors that regulates the growth of populⁿ.

Q.⁸⁶ → Factors regulating population:-

- Birth rate
- Food
- climate
- competition
- Death rate
- water
- Disease
- Emigration
- Disease

Q:⁸⁷ Meta-population: A group of population of same species which is separated by space - Individual can move from one population to other - Extinction & colonization are the main process that regulate meta-population.

Q:⁸⁸ Population Density:

↳ is the no of ind- in population found in particular area

Q:⁸⁹ Higher population density:-

- May help sexually reproduce individual find mates but it can also lead increase comp. for mates, foods, living space, water, sunlight
- Help to shield some members from predators
- close contact among individual in dense population can increase the transmission of parasites & infectious disease
- Smaller organism live higher population density than larger organism

Q:⁹⁰ Low population density:-

- less comp. for food & living space, water, sunlight & other resources.
- they find diff- to find mates
- less adaptability to env. changes-

Q:⁹¹ Density Independent factors:-

factors that influence the no of animals in population & their effect is not dependant on the density of the population. So, D.I exerts a strong effect on population size.

Q⁹²: Population dispersion: or distribution patterns show the spatial relationship b/w members of population within habitat.

Q⁹³: density independent factors:

- Weather
- Earthquake
- Floods
- Hurricane
- Fire
- Pollution
- climate changes
- Habitat destruction
- other factors of climate
- Late spring
- severe freeze

Q⁹⁴: Factors that influence population density

- Comp.
- Disease
- food
- Predation
- migration
- Parasitism

Q⁹⁵: Population Integrity?

it refers to the independence or discreteness of subpopulations. Therefore, population complexity is the result of trade-off b/w connectivity & integrity.

Q⁹⁶: Impacts of human population?

H.P increases day by day about 74 billion

Increase in population causes an increase of demand/research

It affected the Env. by putting pressure on natural resources

- Habitat destruction
- Habitat fragmentation
- Oil Spillage
- Pollution
- Coral reef destruction
- Soil erosion
- Ozone layer destruction
- Invasive species

- Today my zoo507 paper.
- Mcqs:
- 1. Living and non. Living things includes ---- (invironment).
- 2. Non. Renewable sources such as aluminun and _____.
- 3.comunity of different ____ interact each other.
- 4.organisms are divided into ____ types.
- 5.viola tricolor live in ____ environment.
- 6.ecological niche is important for the study of ____.
- 7.blind indus dolphen belong to ____ specie.
- 8._____process start ecological food web.
- 9.% age of nitrogen in atmosphere?
- 10. _____Factor helpful to regulate the population?
- 11.%age of water that cover the earth surface.
- 12._____ zone contain mountain head water streams.
- 13. Ectotherm organisms regulate their body temperature through_____.
- 14._____ wave length of electromagnetic radiations detect human eye.
- 15. Tropical rain forest contain ____ of total world life. □ 16.competition among same species are called_____.
- 17.commensalism.
- 18.foliose is growth form of _____.
- 19.camouflage.
- 20.grass land biomes is example of _____.
- 21.El Nino effect occurs in _____ years.
- 22.CFCs stand for.
- 23.the %age of plants and animals species on world.
- 24._____ is basic unit of life and responsible for the trait inherited.
- 25.pollutants.
- 26. Competition within a specie member.
- 27.PAR stand for_____.
- 28.physical location of gene are _____.
- 29.mRNA is further translated into _____.
- 30. When extra nucleotide put into DNA is called_____.
- 31.human affected environment _____ %.
- 32.invasive specie are called _____.
- 33.hibernation in large animals known _____.
- 34.aposematic coloration.
- 35.sulphur circulate through _____ cycle.
- Subjective.
- 1.aestivation? 2 marks
- 2.solid waste ? 2marks
- 3.exploitation? 2 marks
- 4.growth forms of lichens? 3 marks □ 5.acid rain? 3 marks.

- 6.three sources of energy?3 marks
- 7. Three process involve in water cycle? 5 marks
- 8.compitition and difference between interspecific and itraspecific .? 5 marks □ 9.
Primary succession with examples? 5 marks.
- 10.factors responsible for genetic diversity of a small popultion? 10 marks

Zoo 507

→ How plants converted light energy into chemical energy?

(all inf) (Specie)

generalist specie

→ Foundation specie e.g

→ concept of specie e.g

diff. ^{blw} Indicator specie → foundation specie

→ keystone specie e.g

→ Non-active specie e.g

→ How specie diversity affect ecosystem e.g

→ How non-native specie affect on environment e.g

→ Diff blw generalist & specialist specie?

specie diversity? Indicator specie?

inf (population)

→ reproduction potential

→ population dynamics → factor of population dynamics

→ population integrity

→ population dispersion

→ character of population

→ population depression

(Environment)

→ effects of human on environment → factor assembly community

→ ecological footprints are growing

→ To list composition of atmosphere in ecosystem

→ Eutrophication renewable resources?

→ Imp of water H₂O?

→ composition of geosphere → earth science

→ Major factors that support life on earth